Dynamics and Reconstruction in the Observable Measure

Patrick S. Mahon

Abstract

Starting from a σ-additive probability measure on the algebra of discriminable events [7], the Rokhlin disintegration theorem forces a conditional regularity kernel and a minimal sufficient factor Q_*, the coarsest reduction carrying full conditional information. Indexing over time, temporal coherence forces the Chapman–Kolmogorov equation, yielding a Markov semigroup $\{\Pi_t\}$ and Koopman–Perron duality; the dynamics of observation and the underlying state space are derived, not assumed. In a measure-preserving system Q_* becomes the delay map Φ_h, and reconstruction — that $\mathcal{O}_h = \mathcal{B} \bmod \mu$, density of the delay algebra in L^2, and injectivity of Φ_h are equivalent — is the faithfulness question about this factor. The Stone space [7] built to derive σ-additivity is identified with X itself, up to measure-theoretic isomorphism, when reconstruction holds.

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1 Introduction

Koopman studied dynamical systems through observables [5], but the dynamical law T remains primitive: the operator $U_T f = f \circ T$ is defined only after an underlying evolution has already been specified. The classical stochastic-process tradition proceeds similarly. Prediction is formulated relative to a filtration, so temporal order and information flow are built into the formalism from the outset [2]. That ordering is natural, but it need not be conceptually prior. One may instead ask whether observable probability already carries a conditional regularity between observables, and whether the familiar semigroup and predictive structures arise only when that regularity is organised coherently over time.

In [7], σ-additivity is not assumed but derived: a probability measure is forced into existence by the coherence of finite observations, at the cost of a single admissibility condition that no first-order structural requirement can supply. The measure is earned, not given. It does, however, already carry more than it was built to carry. Any two observations Q and F — no hypothesis on time or order — determine a conditional regularity of F given Q: the Markov kernel $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$, which the Rokhlin disintegration theorem [8] forces into existence. It is not added; it is already there. Composing with Q gives the minimal sufficient factor Q_*, the coarsest reduction of the sample point that carries full conditional information about F. Temporal prediction is the special case in which F is a future observable; nothing in the construction requires it.

Index the conditional regularity over time and the kernels $\{\Pi_t\}$ must compose consistently. That coherence forces Chapman–Kolmogorov — derived, not assumed — and from it a Markov semigroup and Koopman–Perron duality $\int K_t g \, d\mu = \int g \, d(\mu \star \Pi_t)$. In a measure-preserving system with $Q_t = h \circ T^t$, the factor Q_* becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the question becomes internal: does this factor see all of X, or only a proper quotient? The answer — that $\mathcal{O}_h = \mathcal{B} \bmod \mu$, density of the delay algebra in L^2, and injectivity of Φ_h are all equivalent — is the reconstruction theorem. The Stone space [7] built to derive σ-additivity is identified with X itself when reconstruction holds.

2 Conditional regularity and the canonical factor

2.1 The conditional regularity kernel

Let $(\Omega, \mathcal{F}, P)$ be a σ-additive probability space on an observable algebra [7], and let Q, F be two measurable maps:

- $Q : \Omega \to \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \to \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [8] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β, equipped with its canonical (Giry) σ-algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$-null sets) Markov kernel $\kappa_Q : \alpha \to \mathcal{P}(\beta)$ satisfying

$$\int_A \kappa_Q(q, B)\, d(P \circ Q^{-1})(q) \;=\; P(Q \in A,\ F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Conditional regularity kernel)**.** The *conditional regularity kernel* of F given Q is κ_Q, the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\kappa_Q(q, A) = P(F \in A \mid Q = q)$. When F is a future observable the kernel describes conditional prediction; the object itself is neutral on time.

The kernel is covariant under measurable maps: if $\pi : \alpha \to \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\kappa_Q(q, \cdot) = \kappa_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$-almost every q, by uniqueness of disintegration.

Composing κ_Q with Q gives the map sending each sample point to its conditional regularity.

2.2 The minimal sufficient factor

Definition 2.2 (Minimal sufficient factor)**.** The *minimal sufficient factor* is

$$Q_* : \Omega \to \mathcal{P}(\beta), \qquad Q_*(\omega) \;=\; \kappa_Q(Q(\omega), \cdot).$$

Since κ_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Factorization via the canonical factor)**.** *For every bounded measurable $g : \beta \to \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] \;=\; \int g\, dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g\, dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g\, d\kappa_Q(q, \cdot)$ is strongly measurable; since $Q_* = \kappa_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$-measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned}\int_S h\,dP &= \int_A \int g\,d\kappa_Q(q,\cdot)\,d(P\circ Q^{-1})(q) \quad \text{(pushforward)}\\ &= \int_{A\times\beta} g(f)\,d\rho(q,f) \quad \text{(disintegration)}\\ &= \int_S g(F)\,dP.\end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. □

Since $\sigma(Q_*) \subseteq \sigma(Q)$ and the right-hand side is already $\sigma(Q_*)$-measurable, Q_* is a sufficient statistic for F given Q: $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P-a.e. It is *minimal* in the sense that every sufficient statistic factors through it: Q_* is the coarsest reduction of ω that carries the full conditional regularity of F given Q. The operator $(K_{Q_*}g)(q_*) = \int g\,dq_*$ on $\mathcal{P}(\beta)$ recasts this as $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g)\circ Q_*$ P-a.e.

3 Dynamics and duality

3.1 Markov semigroup

When the conditional regularity kernel κ_Q is indexed over a time parameter, the kernels cannot be chosen independently: the regularity of F at time $t+s$ given Q at time 0 must be consistent with first observing at time s and then at time t. This consistency requirement forces the Chapman–Kolmogorov equation.

Concretely, index over a discrete time horizon $\mathbb{N}$ and let $\Pi_t : \gamma \to \mathcal{P}(\gamma)$ be a Markov kernel on a state space γ, with $\Pi_0(q,\cdot) = \delta_q$. Temporal coherence of the conditional regularity demands

$$\Pi_{t+s}(q,\cdot) \;=\; \int_\gamma \Pi_t(q',\cdot)\,d\Pi_s(q,q') \quad \text{(i.e., } \Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s\text{)}$$

— the Chapman–Kolmogorov equation, derived from the structure rather than assumed. A family $\{\Pi_t\}$ satisfying it is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) \;:=\; \int g\,d\Pi_t(q,\cdot), \qquad g : \gamma \to \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \mathrm{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 3.1 (Semigroup property)**.** *$K_{t+s}g = K_t(K_s g)$ for all bounded measurable g.*

Proof. $(K_{t+s}g)(q) = \int g\,d(\Pi_t \circ_\kappa \Pi_s)(q,\cdot) = \int\left(\int g\,d\Pi_t(q',\cdot)\right)d\Pi_s(q,q') = \int (K_t g)(q')\,d\Pi_s(q,q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. □

The pushforward of a measure μ on γ through the kernel is $P_t^*\mu := \mu \star \Pi_t = \int \Pi_t(q,\cdot)\,d\mu(q)$.

3.2 Koopman–Perron duality

Theorem 3.2 (Koopman–Perron duality)**.** *For every bounded measurable g and finite measure μ:*

$$\int (K_t g)\,d\mu \;=\; \int g\,d(P_t^*\mu).$$

Proof. $\int g\,d(P_t^*\mu) = \int g\,d(\mu\star\Pi_t) = \int_\gamma \int g\,d\Pi_t(q,\cdot)\,d\mu(q) = \int (K_t g)\,d\mu$, by Bochner–Fubini. □

When $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \to \gamma$, the stochastic picture collapses.

3.3 Specialisation to measure-preserving systems

Proposition 3.3 (Deterministic limit)**.** *If* $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$*, then* $K_t g = g \circ \varphi_t$ *and* $\varphi_{t+s} = \varphi_t \circ \varphi_s$*.*

Proof. $(K_t g)(q) = \int g\, d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q,\cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. □

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g\, d\mu = \int g\, d(P_t^* \mu)$ becomes the change-of-variables formula $\int g \circ T^t\, d\mu = \int g\, d(T_*^t \mu)$.

Example 3.4 (Finite Markov chain)**.** Let $X = \{1,\dots,k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution μ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The conditional regularity kernel of X_1 given X_0 is $\kappa(i,\cdot) = P(i,\cdot)$, the i-th row of P: this is the unique Markov kernel satisfying $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$ for all bounded f. The t-step kernel $\Pi_t(i,\cdot) = P^t(i,\cdot)$ is the corresponding semigroup kernel, and temporal coherence gives $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g\, d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g\, d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically σ-additive, so the admissibility condition of [7] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever h separates points of X.

The full structure produced by the construction is the triple $(\mathcal{P}(\beta),\, \nu_{Q_*},\, \{K_t\}_{t\in\mathbb{N}})$: the canonical factor space, stationary distribution $\nu_{Q_*} = P \circ Q_*^{-1}$, and Markov semigroup. This is the *observable dynamical system* derived from (Q, F, P): the conditional regularity of F relative to Q, promoted to a dynamical object by temporal coherence.

4 From regularity to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n\omega))_{n\in\mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n\in\mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The canonical factor space $\Phi_h(X)$ is a subset of $\mathbb{R}^\mathbb{Z}$ encoding the dynamical information that h can see. Whether it is a faithful image of X depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X. Injectivity of Φ_h modulo μ-null sets is equivalent to the observable σ-algebra

$$\mathcal{O}_h = \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big)$$

equalling $\mathcal{B}(X)$ modulo μ.

Definition 4.1 (Faithful factor)**.** A measurable map $\psi : (X, \mathcal{B}, \mu) \to (Y, \mathcal{Y})$ is *faithful modulo* μ if

$$\sigma(\psi) = \mathcal{B} \quad \text{modulo } \mu,$$

equivalently, if ψ separates μ-a.e. pair of points of X. The reconstruction question is whether Φ_h is faithful modulo μ.

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable σ-algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$.

4.1 Separation defect and the role of CE

For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$ from a directed system $(\mathcal{E}_n)_n$, the *separation defect* is

$$\delta(\mathcal{G}) \;:=\; \sup_{A\in\sigma(\mathcal{C})} \inf_{E\in\mathcal{G}} \mu(A\triangle E),$$

where $\sigma(\mathcal{C})$ is the observable σ-algebra of the full system. It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C})$ mod μ. A sequence $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C})$ mod μ; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Theorem 4.2 (CE drives separation defect to zero)**.** *Let $(\mathcal{E}_n)_n$ be a query system with realization measure μ satisfying collective exhaustion [7]. Under any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \;\longrightarrow\; 0 \quad \text{as } k \to \infty.$$

Proof. Fix $A \in \sigma(\mathcal{C})$. Since $(\mathcal{G}_k)$ is honest, $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C})$ mod μ, so A is $\sigma(\bigcup_k \mathcal{G}_k)$-measurable. By the $L^1(\mu)$ martingale convergence theorem, $\mathbf{1}_{E_k} \to \mathbf{1}_A$ in $L^1(\mu)$ where $E_k = \mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$ rounded to the nearest atom; in particular $\mu(A\triangle E_k) \to 0$. CE [7] ensures no mass persists on the vanishing sets $A\triangle E_k$, confirming $\inf_{E\in\mathcal{G}_k} \mu(A\triangle E) \to 0$ uniformly over A. Hence $\delta(\mathcal{G}_k) \to 0$. $\square$

Sections 5–6 carry out the reconstruction in the delay-map setting.

Remark 4.3 (The temporal specialisation)**.** With $Q_t = h \circ T^t$, the algebra $\mathcal{O}_h^{(L)}$ is an honest refinement scheme once reconstruction holds (§§5–6). The separation defect specialises to $\delta(L) := \delta(\mathcal{O}_h^{(L)})$, the central quantity of [6]: exponential decay of $\delta(L)$ under mixing yields minimax-optimal certification rates. In the general query setting, exhaustion and mixing are separate conditions; the delay-map setting collapses the first by construction.

5 Density bridge and reconstruction

5.1 Density bridge lemma

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \to X$ is a measurable, μ-preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 5.1 (Observable algebra)**.** The *observable algebra* generated by h is

$$\mathcal{O}_h \;:=\; \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big) \;\subseteq\; \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X,\mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 5.2 (Delay map)**.** The *delay map* is

$$\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}, \qquad \Phi_h(x) \;=\; \big(h(T^n x)\big)_{n\in\mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n, the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B}$ mod μ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞.

Lemma 5.3 (Density bridge)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function* 1*. Then:*

$$\overline{\mathcal{A}}^{L^2} \;=\; L^2(X,\, \sigma(\mathcal{A}),\, \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ.

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$-measurable, hence $\mathcal{M}$-measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V. By the functional monotone class theorem [3], the class

$$\mathcal{H} \;:=\; \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \to f$ pointwise a.e. with $f \in L^\infty$, then $f_n \to f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$-indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$-simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B}$ mod μ. □

Remark 5.4 (Role of the algebra structure)**.** The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear* span $\mathcal{H}_{\mathcal{A}} = \overline{\operatorname{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu =$ Lebesgue, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ-algebra (the dyadic structure separates points). But $\operatorname{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2. So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B}$ mod μ — see Corollary 6.2.

5.2 Reconstruction theorem

Theorem 5.5 (Reconstruction theorem)**.** *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

(i) $\mathcal{O}_h = \mathcal{B}$ modulo μ;

(ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;

(iii) $\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ modulo μ.

Proof. (i) ⇔ (ii). Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B}$ mod μ.

(i) ⇔ (iii). By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. So $\mathcal{O}_h = \mathcal{B}$ mod μ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B}$ mod μ, which is exactly condition (iii). □

Condition (iii) means Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h$
Since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$, the measure $(\Phi_h)_*\mu$ is shift-invariant, and (X, T, μ) is isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to $\Phi_h(X)$.

6 Cyclic vectors and Stone space identification

6.1 Cyclic vectors

Definition 6.1 (Cyclic vector)**.** The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h \ := \ \overline{\operatorname{span}}\{U_T^n h : n \in \mathbb{Z}\} \ = \ L^2(X, \mu).$$

Corollary 6.2 (Cyclic implies reconstruction)**.** *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T, then $\mathcal{O}_h = \mathcal{B}$ modulo μ.*

Proof. Since $\mathcal{A}_h$ contains $\operatorname{span}\{h \circ T^n\} = \operatorname{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. □

Remark 6.3 (The implication is strict)**.** The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h, the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2-spectrum*: $L^2(X, \mu)$ is a cyclic U_T-module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [4] but is not universal.

6.2 Stone space identification

The Stone space $\mathrm{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \to \mathrm{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Theorem 6.4 (Stone space identification)**.** *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ. Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\mathrm{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\mathrm{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. Injectivity mod μ. Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\ni x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ-a.e.

σ-algebra identification. The pullback ι^{-1} maps the Baire σ-algebra of $\mathrm{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

Measure recovery. Since μ is σ-additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [7] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. □

The Stone space [7] — built as a compact ambient space for deriving σ-additivity — is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

Example 6.5 (Cantor set: reconstruction without smoothness)**.** Let $\mathcal{C} \subset [0, 1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \ldots) = \sum_{k \geq 1} d_k/3^k$. Let $S : \mathcal{C} \to \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0,2\}^{\mathbb{N}}, \nu, \sigma)$ via π; in particular S is $\mu_{\mathcal{C}}$-invariant and ergodic.

For $x = \pi(d_1, d_2, \ldots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \ldots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in \left[0, \tfrac{1}{3}\right], \qquad d_{n+1} = 2 \iff S^n x \in \left[\tfrac{2}{3}, 1\right],$$

so $h \circ S^n$ recovers the $(n+1)$-st ternary digit of x. The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \ldots, S^L x)$ reads off digits $d_1, \ldots, d_{L+1}$; its fibres are the generation-$(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k,\ 1 \le k \le L+1\}, \quad z \in \{0,2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0,2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$-a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \to \infty$:

$$\mathcal{O}_h \;=\; \sigma\Big(\bigcup_{L \ge 0} \mathcal{O}_h^{(L)}\Big) \;=\; \mathcal{B}(\mathcal{C}) \mod \mu_{\mathcal{C}}.$$

By Theorem 5.5, Φ_h is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \to \infty$.

The observable algebra also encodes the geometry of $\mathcal{C}$ directly. At each lag L there are 2^{L+1} fibres of Euclidean diameter $(1/3)^{L+1}$, so the box-counting ratio $\log 2^{L+1}/\log 3^{L+1} = \log 2/\log 3$ is an algebraic invariant of the algebra itself, constant across all lags. The Hausdorff dimension is not a separate geometric calculation: it is read off from the refinement rate of the observable algebra.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2/\log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — C^2 ambient dynamics, integer manifold dimension d, embedding bound $2d+1$ — are unavailable before one even asks about regularity of S. Takens has no domain of application here. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness plays no role.

The classical Takens embedding theorem [9] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d-manifold and a generic smooth h, the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \ldots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \ge 2d+1$. The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h, the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2-dense: the density condition holds and the two results are compatible. For non-smooth or zero-dimensional systems — the Cantor example being representative — the closer historical ancestor is the symbolic dynamics and generating-partition tradition [1]: representing orbits through successively finer finite partitions is exactly what the observable algebra $\mathcal{O}_h^{(L)}$ does as L grows, with faithfulness of Φ_h replacing the generating-partition condition.

Remark 6.6 (Observable separation and Lyapunov exponents)**.** When T is C^r and reconstruction holds, the delay map Φ_h carries a natural instability rate: for μ-a.e. x and nearby y,

$$\lambda_h(x, y) \;:=\; \limsup_{n \to \infty} \frac{1}{n} \log d\big(\Phi_h(T^n x),\ \Phi_h(T^n y)\big).$$

This *observable separation exponent* measures how fast the reconstructed dynamics distinguishes initial conditions through the observable, rather than how fast tangent vectors grow. When h is generic in the sense of Takens [9], the finite-lag embedding $\Phi_h^{(L)}$ is bi-Lipschitz on the attractor, sandwiching λ_h between the top Lyapunov exponent λ_1 from above and below; one expects $\lambda_h = \lambda_1$ μ-a.e., but making this precise requires comparing the induced metric on $\Phi_h^{(L)}(X)$ with the ambient Riemannian geometry as $L \to \infty$, which is beyond the scope of the present paper. The quantitative geometry of fibre separation in this setting is taken up in [6].

7 Discussion

The measure derived in [7] already contains, without further assumption, the structures developed here. The Rokhlin disintegration reads off a conditional regularity kernel for any pair of observables; temporal indexing of that kernel forces Chapman–Kolmogorov and hence the Markov semigroup; and in a measure-preserving system the minimal sufficient factor becomes the delay map. Nothing is added to the measure — only consequences are drawn.

The reconstruction theorem closes a circle opened in [7]. The Stone space $\mathrm{St}(\mathcal{C})$ built there to establish σ-additivity is an abstract completion of the observable algebra, containing the realised state space as the principal ultrafilters but not yet identified with it. When Φ_h is faithful modulo μ (Definition 4.1), the three equivalent conditions of Theorem 5.5 jointly force $\sigma(\Phi_h) = \mathcal{B}$ modulo μ, and the Stone space collapses onto X itself: the abstract completion is the concrete state space. The question left open in [7] — whether prediction, dynamics, and state recovery follow from observation — is answered affirmatively, under faithfulness.

What faithfulness means in practice, and whether it can be certified from finite data, is taken up in [6].

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